Problem Set 4

Directions

Submit two files on Moodle before the deadline:
[1] your **solution set** as a *.pdf [2] the **R script** as a *.R type file

For both, the first line should include your student ID and the assignment number
If you worked with one or two other students write "group:" and then their student ID numbers
[1] the **solution set** should include for each answer:

- the question number/part
- your answer in words
- the command in R that was used (if relevant)
- the output from Rstudio (if relevant)

[2] the **R script** should include:

- all the commands you used to generate the solutions
- individually written comments that explain your code to the grader

Primary seatbelt laws and traffic fatalities

A primary seat belt law gives law enforcement the right to ticket a driver for not wearing a seatbelt, regardless of whether they have broken any other laws such as speeding. This is as opposed to a secondary seatbelt law which allows ticketing for any secondary passengers that are not wearing their seatbelt. In the USA, between 1981 and 2003 different states implemented such laws at different times.

You have access to a panel dataset from Anderson (2008) saved as **traffic.data** in the *.Rdta data file on moodle.

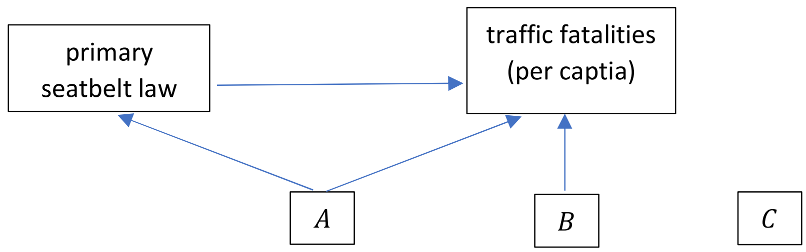
- **state** a number representing each of the 48 states in contiguous US
- **year** sample year, 1981 – 2003

For each state i in each year t we observe the following variables:

- **beer**: per capita beer consumption in the state (gallons of ethanol)
- **primary**: indicator that the state has a primary seat belt law in year t. For a given state the variable primary is 0 until the year the seatbelt law passes, after which it will be 1 for every year after that.
- **secondary**: indicator that the state has a secondary seat belt law
- **totalvmt**: vehicle miles traveled (VMT) in the state
- **precip**: annual precipitation (inches) (e.g. rain)
- **snow32**: annual snow (inches)
- **rural_speed**: rural interstate highway speed limit (mi/hr)
- **urban_speed**: urban interstate highway speed limit (mi/hr)
- **ln_fatal_pc**: log of traffic fatalities per capita (this is your outcome variable)

Research question: **What is the effect of the primary seatbelt law on traffic fatalities per capita?**

1. Causal Diagram. Consider the diagram below. Categorize the following variables as belonging to A, B, or C: **beer**, **secondary**, **totalvmt**, **precip**, **snow32**, **rural_speed**. Explain your reasoning for each variable



2. Pooled OLS. Use the dataset **traffic.data** (all observations from 1981 – 2003) and estimate a pooled OLS model that ignores the panel structure of the data.

a. Write down the model you decide to use (call it "Pooled"). State explicitly which coefficient will you use to answer your research question, and discuss which control variables you include (refer to your answer in Question 1).

b. Estimate the model in R and report your results. Why might this estimate be biased?

3. Fixed effects. Consider the following fixed effects model (FE1):

$$\log(fatal.pc_{it}) = \alpha_i + \beta_1 primary_{it} + \beta_2 year_t + \boldsymbol{\gamma}' X_{it} + u_{it}$$

The term X_{it} captures all the control variables you chose to include (use the same ones as in Question 2).

a. What does α_i capture? How do we interpret β_1? How do we interpret β_2?

b. Estimate FE1 using clustered standard errors <u>at the state level</u>. Why is it appropriate to cluster at the state level?

c. Estimate the model and report your results. What can you conclude regarding the research question?

4. Two-way fixed effects. Consider the model (FE2) with both state and year fixed effects:

$$\log(fatal.pc_{it}) = \alpha_i + \lambda_t + \beta_1 primary_{it} + \boldsymbol{\gamma}' X_{it} + u_{it}$$

a. What does λ_t capture? Why might we prefer (FE2) over the linear time trend in (FE1)?

b. Estimate FE2 using clustered standard errors <u>at the state level</u> and report your results. What can you conclude regarding the research question?